

THE STRUCTURE OF MATHEMATICAL REALITY

Phase V — Chapter 17

Open Questions and the Shape of the Unknown

Assignment

Self-Study Curriculum

Produce three things:

1. **The final diagram.** Revise your map of the mathematical fields one last time. It should now include: the fields, the bridges (as functors), the hierarchy of structure preservation, at least three concepts from your own work placed on the map, and the open questions positioned at the boundaries of the map where they belong.
2. **A one-page statement of your current understanding.** Answer the question: “What is mathematics, what does it do when it works, and why does it connect to physical reality?” This should be written in your own voice, using the vocabulary you have acquired. It will not be complete. It should be honest about where it is incomplete.
3. **A personal research agenda.** List 3-5 questions that you want to answer next. For each, state: what you would need to learn, where you would start, and what a satisfying answer would look like. These are not assignments, they are directions. The purpose of this curriculum was to give you a map. The agenda is where you choose to go.